

Fig. 4: Group of pictures

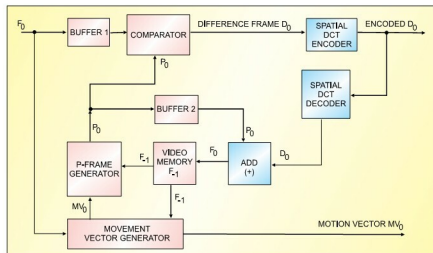


Fig. 5: Temporal predictions

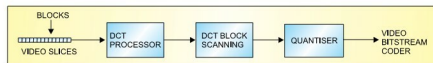


Fig. 6: Discrete cosine transform

known as the inter-frame (I-frame). This is followed by a P-frame obtained by comparing the second frame with the I-frame. This is then repeated and the third frame is compared with the previous P-frame to produce a second P-frame and so on. This goes on until the end of the group of twelve frames. A new reference I-frame is then inserted for the next group of twelve frames and so on. This type of prediction is known as forward prediction.

Motion vector. Motion vector is obtained by the process of block matching. Y component of the reference frame is divided into 16x16 macro blocks. Blocks are taken one by one and the matching block searched within the given area.

When a match is found, the displacement is used to obtain a motion-compensation vector that describes the movement of the macro block in terms of speed and direction. Only a relatively small amount of data is necessary to describe a motion-compensation vector. The actual pixel values of the macro block do not have to be retransmitted.

Predicted and difference frames. As shown in Fig. 5, current frame F_0 is fed into Buffer-1 and held there for a while. It is also fed into the movement vector generator, which uses the contents of the previous frame F_{-1} stored in the video memory to obtain motion vector MV_0 . The motion vector is then added to F_{-1} to produce predicted frame P_0 . P_0 is compared with

the contents of the current frame F_0 in Buffer-2 to produce residual error or difference frame D_0 . Residual error D_0 is fed into the spatial discrete cosine transform (DCT) encoder and sent out for transmission.

Encoded D_0 is decoded to reproduce D_0 as it would be required at the receiving end. D_0 is then added to P_0 , which has been waiting in Buffer-2 to reconstruct current frame F_0 for storage in the video memory for the next frame and so on.

Spatial compression

The heart of spatial redundancy removal is the DCT processor. The DCT processor receives video slices in the form of a stream of 8x8 blocks. The blocks may be part of a luminance frame (Y) or a chrominance frame. Sample values representing the pixel of each block are then fed into the DCT processor, which translates them into an 8x8 matrix of DCT coefficients representing the spatial frequency content of the block. The coefficients are then scanned and quantised before transmission.

Discrete cosine transform. DCT is used in several coding standards (MPEG 1, MPEG 2, H.263, etc) to remove spatial redundancy that exists between neighbouring pixels in a image. It is a Fourier transform which takes information in the time domain and expresses it in the frequency domain. Normal pictures are two-dimensional and contain diagonal as well as horizontal and vertical spatial frequencies. MPEG-2 specifies DCT as the method of transforming spatial picture information into spatial frequency components. Each spatial frequency is given a value known as the DCT coefficient. For an 8x8 block of pixel samples, an 8x8 block of DCT coefficients is produced (refer Fig. 7).

A block that contains different picture details is represented by various coefficient values in the appropriate cells. Coarse picture details utilise a number of cells towards the left top